Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A steel wire of length l and cross sectional area A is stretched by $1\ \mathrm{cm}$ under a given load. When the same load is applied to another steel wire of double its length and half of its cross section area. Then what is the amount of stretching (extension) in cm ?

(A) 3

(B) 5

(C) 6

(D) 4

Q.2

Two wires A and B of same material and of equal length with the radii in the ratio $1:2$ are subjected to identical loads. If the length of A increases by $8\ \mathrm{mm}$, then what will be the increase in length (in mm) of B ?

(A) 1

(B) 3

(C) 4

(D) 2

Q.3

What is the minimum diameter of a brass rod if it is to support a $400\ \mathrm{N}$ load without exceeding the elastic limit? Assume that the stress for the elastic limit is $379\mathrm{MPa}$.

(A) $1.16\ \mathrm{mm}$

(B) $2.32\ \mathrm{mm}$

(C) $0.16\ \mathrm{mm}$

(D) $1.35\ \mathrm{mm}$

Q.4

The elastic limit of steel is $8 \times 10^8\ \mathrm{N/m^2}$ and its Young's modulus $2 \times 10^{11}\ \mathrm{N/m^2}$. Find the maximum elongation of a half-meter steel wire that can be given without exceeding the elastic limit.

(A) $2\ \mathrm{mm}$

(B) $4\ \mathrm{mm}$

(C) $5\ \mathrm{mm}$

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(D) $6\ \text{mm}$

Q.5

Two wires A and B of same material and of equal length with the radii in the ratio $1:2$ are subjected to identical loads. If the length of A increases by $8\ \text{mm}$, then the increase in length of B is

(A) $2\ \text{mm}$

(B) $4\ \text{mm}$

(C) $8\ \text{mm}$

(D) $16\ \text{mm}$

Q.6

The length of elastic string, obeying Hooke's law is ℓ_1 metres when the tension $4\ \text{N}$ and ℓ_2 metres when the tension is $5\ \text{N}$. The length in metres when the tension is $9\ \text{N}$ is -

(A) $5\ell_1 - 4\ell_2$

(B) $5\ell_2 - 4\ell_1$

(C) $9\ell_1^1 - 8\ell_2^2$

(D) $9\ell_2^2 - 8\ell_1^1$

Q.7

A steel wire of length l and cross section area A is stretched by $1\ \text{cm}$ under a given load. When the same load is applied to another steel wire of double its length and half of its cross section area, the amount of stretching (extension) is

(A) $0.5\ \text{cm}$

(B) $2\ \text{cm}$

(C) $4\ \text{cm}$

(D) $1.5\ \text{cm}$

Q.8

A force of $6 \times 10^6 \text{Nm}^{-2}$ is required for breaking a material. Then density ρ of the material is $3 \times 10^3\ \text{kgm}^{-3}$. If the wire is to break under its own weight, the length of the wire made of that material should be (take $g = 10\ \text{ms}^{-2}$)

(A) $20\ \text{m}$

(B) $200\ \text{m}$

(C) $100\ \text{m}$

(D) $2000\ \text{m}$

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Q.9

A steel wire of cross-sectional area $3 \times 10^{-6}\ \mathrm{m}^2$ can with stand a maximum strain of 10^{-3}. Young's modulus of steel is $2 \times 10^{11}\ \mathrm{N/m^2}$. The maximum mass the wire can hold is:

(A) $40\ \mathrm{kg}$

(B) $60\ \mathrm{kg}$

(C) $80\ \mathrm{kg}$

(D) $100\ \mathrm{kg}$

Q.10

A structural steel rod has a radius of $10\ \mathrm{mm}$ and length of $1.0\ \mathrm{m}$. A $100\mathrm{kN}$ force stretches it along its length. Young's modulus of structural steel is $2 \times 10^{11}\mathrm{Nm}^{-2}$. The percentage strain is about

(A) 0.16%

(B) 0.32%

(C) 0.08%

(D) 0.24%

Q.11

A thick rope of density ρ and length L is hung from a rigid support. The Young's modulus of the material of rope is Y. The increase in length of the rope due to its own weight is

(A) $(1/4)\rho g L^2/Y$

(B) $(1/2)\rho g L^2/Y$

(C) $\rho g L^2/Y$

(D) $\rho g L/Y$

Q.12

A rubber cord catapult has cross-sectional area $25\ \mathrm{mm}^2$ and initial length of rubber cord is $10\ \mathrm{cm}$. It is stretched to $5\ \mathrm{cm}$ and then released to project missile of mass $5\mathrm{gm}$. Taking $\mathrm{Y}_{\text{pubber}} = 5 \times 10^8\ \mathrm{N/m^2}$ Velocity of projected missile is

(A) $20\ \mathrm{ms}^{-1}$

(B) $100\ \mathrm{ms}^{-1}$

(C) $250\ \mathrm{ms}^{-1}$

(D) $200\ \mathrm{ms}^{-1}$

Q.13

A steel wire $1.5\ \mathrm{m}$ long and of radius $1\ \mathrm{mm}$ is attached with a load $3\ \mathrm{kg}$ at one end the other end of the wire is

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fixed. It is whirled in a vertical circle with a frequency $2\ \text{Hz}$. Find the elongation of the wire when the weight is at the lowest position. $\left[\text{Y} = 2 \times 10^{11}\ \text{N/m}^2,\ \text{g} = 10\ \text{ms}^{-2}\right]$

(A) $1.77 \times 10^{-3}\ \text{m}$

(B) $7.17 \times 10^{-3}\ \text{m}$

(C) $3.17 \times 10^{-7}\ \text{m}$

(D) $1.37 \times 10^{-7}\ \text{m}$

Q.14

If the ratio of radii of two wires of same material is $2:1$ and ratio of their lengths is $4:1$, then the ratio of the normal forces that will produce the same extension in the length of two wires is

(A) $2:1$

(B) $4:1$

(C) $1:4$

(D) $1:1$

Q.15

A copper wire of length $1.0\ \text{m}$ and a steel wire of length $0.5\ \text{m}$ having equal cross-sectional areas are joined end to end. The composite wire is stretched by a certain load which stretches the copper wire by $1\ \text{mm}$. If the Young's modulii of copper and steel are respectively $1.0 \times 10^{11}\text{Nm}^{-2}$ and $2.0 \times 10^{11}\text{Nm}^{-2}$, the total extension of the composite wire is :

(A) $1.75\ \text{mm}$

(B) $2.0\ \text{mm}$

(C) $1.50\ \text{mm}$

(D) $1.25\ \text{mm}$

Q.16

Arubber cord of mass M, length L and cross-sectional area A is hung vertically from a ceiling. The Young's modulus of rubber is Y. If the change in the diameter of the cord due to its own weight is neglected, the increase in its length due to its own weight is

(A) $\frac{MgL}{AY}$

(B) $\frac{MgL}{2AY}$

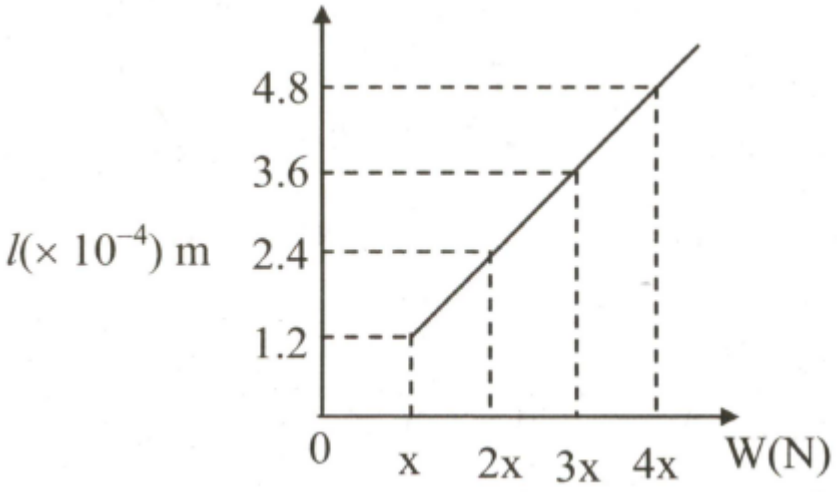
(C) $\frac{2MgL}{AY}$

(D) $\frac{MgL}{4AY}$

Q.17

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Consider a 1 m long steel wire of cross-sectional area 10^{-6} m^2 is suspended from rigid support and a load is attached at the other end. In the below figure, the graph of extension in the wire versus load is shown:



If graphical calculation gives the value of Young's modulus as 2×10^{11} N / m^2, then find the value x (in newton):

(A) 25.00

(B) 26.00

(C) 23.00

(D) 24.00

Q.18

Energy per unit volume in a stretched wire is

(A) Half of load $\times$ strain

(B) Load $\times$ strain

(C) Stress $\times$ strain

(D) Half of stress $\times$ strain

Q.19

A load of 40 kg is suspended from a wire of radius 10^{-3} m and density 8×10^3 kgm^{-3} . 75% of the work done to increase the length is converted into heat. Calculate the change in temperature (in kelvin) of the wire . Given that $Y = 7.5 \times 10^{10}$ Nm^{-2} and heat capacity of the material of the wire is 490 $Jkg^{-1}K^{-1}$ respectively. (Take $\pi^2 = 9.8$ and $g = 9.8$ m / s^2)

(A) 0.06

(B) 0.09

(C) 0.04

(D) 0.02

Q.20

Length of an elastic string is x metre when longitudinal tension is 3 N and y metre when it is 5 N. If length of string is found to be 2.5 $y - 1.5$ x, calculate tension in the string in N.

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(A) 7

(B) 9

(C) 5

(D) 8

Answers & Solutions

Q.1 Answer:

4

Solution:

Young's modulus of elasticity is

$$Y = \frac{F/A}{\Delta L/L}$$

$$\therefore \Delta L = \frac{FL}{AY}$$

So, $\Delta L \propto \frac{L}{A}$

$$\therefore \frac{\Delta L_2}{\Delta L_1} = \frac{L_2}{L_1} \times \frac{A_1}{A_2} = \frac{2}{1} \times \frac{2}{1} = 4$$

$$\Delta L_2 = 4 \times \Delta L_1 = 4 \times 1 = 4 \text{ cm}$$

Q.2 Answer:

2

Solution:

Ratio of radii $r_1 : r_2 = 1 : 2$

Ratio of area, $A_1 : A_2 = \pi r_1^2 : \pi r_2^2$

$$A_1 : A_2 = 1 : 4$$

Now, Stress $_1$ **: Stress** $_2 = 4 : 1$ **So, Strain** $_1$ **: Strain** $_2 = 4 : 1$

$$\therefore \frac{l_1}{l_2} = \frac{4}{1} \Rightarrow 4l_2 = l_1 = 8$$

$$\therefore \quad l_2 = 2 \text{ mm}$$

Increase in length of B **is** 2 **mm.**

Q.3 Answer:

1.16 mm

Solution:

To find the minimum diameter, and hence minimum cross-sectional area, we assume that the force $F = 400 \text{ N}$ **brings us to the elastic limit. Then from the stress,** $F/A = 379 \times 10^6 \text{ Pa}$**, we get**

$$A = \frac{400 \text{ N}}{379 \times 10^6 \text{ Pa}} = 1.0554 \times 10^{-6} \text{ m}^2$$

Then, $A = \frac{\pi D^2}{4}$

$$D^2 = \frac{4A}{\pi} = \frac{4\left(1.0554 \times 10^{-6} \text{ m}^2\right)}{\pi}$$

$$= 1.344 \times 10^{-6}$$

and $D = \sqrt{1.344 \times 10^{-6} \text{ m}^2}$

$$= 1.16 \times 10^{-3} \text{ m} = 1.16\text{mr}$$

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Q.4 Answer:

2 mm

Solution:

$$Y = \left(\frac{F}{A}\right) \times \frac{L}{\Delta L}$$

$$\Rightarrow \Delta L \left(\frac{F}{A}\right) \frac{L}{Y} = \frac{8 \times 10^8 \times 0.5}{2 \times 10^{11}} = 2 \text{ mm}$$

Q.5 Answer:

2 mm

Solution:

Ratio of radii $r_1 : r_2 = 1 : 2$ **Ratio of area,** $A_1 : A_2 = \pi r_1^2 : \pi r_2^2$

$A_1 : A_2 = 1 : 4$

Now, Stress $_1$ **: Stress** $_2 = 4 : 1$ **So, Strain** $_1$: $\text{Strain}_2 = 4 : 1$

$\therefore \quad \frac{l_1}{l_2} = \frac{4}{1} \Rightarrow 4l_2 = l_1 = 8$

$\therefore \quad l_2 = 2 \text{ mm}$

Increase in length of B **is** 2 mm.

Q.6 Answer:

$5\ell_2 - 4\ell_1$

Solution:

Let ℓ_0 **be the unstretched length and** ℓ_3 **be the length under a tension of** 9 N. **Then** $Y = \frac{4\ell_0}{A(\ell_1 - \ell_0)} = \frac{5\ell_0}{A(\ell_2 - \ell_0)} = \frac{9\ell_0}{A(\ell_3 - \ell_0)}$

These give

$\frac{4}{\ell_1 - \ell_0} = \frac{5}{\ell_2 - \ell_0} \Rightarrow \ell_0 = 5\ell_1 - 4\ell_2$

Further, $\frac{4}{\ell_1 - \ell_0} = \frac{9}{\ell_2 - \ell_0}$

Substituting the value of ℓ_0 **and solving,**

we get $\ell_3 = 5\ell_2 - 4\ell_1$

Q.7 Answer:

4 cm

Solution:

Young's modulus of elasticity is $Y = \frac{F/A}{\Delta L/L} \quad \therefore \Delta L = \frac{FL}{AY}$

So, $\Delta L \propto \frac{L}{A}$

$\therefore \frac{\Delta L_2}{\Delta L_1} = \frac{L_2}{L_1} \times \frac{A_1}{A_2} = \frac{2}{1} \times \frac{2}{1} = 4$

$\Delta L_2 = 4 \times \Delta L_1 = 4 \times 1 = 4 \text{ cm}$

Q.8 Answer:

200 m

Solution:

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Breaking stress $= \frac{\text{Force}}{\text{area}}$ **The breaking force will be its own weight.** $\mathrm{F} = \mathrm{mg} = \mathrm{V}\rho\mathrm{g} =$ **area** $\times \ell\rho\mathrm{g}$
Breaking stress $= 6 \times 10^6 = \frac{\text{area} \times \ell \times \rho g}{\text{area}}$
or $\ell = \frac{6\times10^6}{3\times10^3\times10} = 200$ m.

Q.9 Answer:

60 kg

Solution:

$A = 3 \times 10^{-6}$
$\frac{\Delta L}{L} = 10^{-3}$
$\gamma = 2 \times 10^{11}$
$f = mg$
$Y = \frac{FL}{A\Delta L}$
$\gamma = \frac{mg}{3\times10^{-6}\times10^{-3}}$
$2 \times 1^{11} = \frac{m\times10}{3\times10^{-9}}$
$2 \times 10^{10} \times 3 \times 10^{-9} = m$
$m = 60$ kg

Q.10 Answer:

0.16%

Solution:

Given: $F = 100\mathrm{kN} = 10^5$ N
$\mathrm{Y} = 2 \times 10^{11}\mathrm{Nm}^{-2}$
$\ell_0 = 1.0$ m
radius $\mathrm{r} = 10$ mm $= 10^{-2}$ m
From formula, $Y = \frac{\text{Stress}}{\text{Strain}} \Rightarrow$ **Strain** $= \frac{\text{Stress}}{Y} = \frac{F}{AY}$
$= \frac{10^5}{\pi r^2 Y} = \frac{10^5}{3.14\times10^{-4}\times2\times10^{11}} = \frac{1}{628}$
Therefore % **strain** $= \frac{1}{628} \times 100 = 0.16\%$

Q.11 Answer:

$(1/2)\rho g L^2/Y$

Solution:

$$\Delta\ell = \frac{F(L/2)}{AY} = \frac{(AL\rho g)(L/2)}{AY}$$
$$= \left(\frac{1}{2}\right)\rho g L^2/Y$$

Q.12 Answer:

250 ms^{-1}

Solution:

Young's modulus of rubber, $Y_{\text{rubber}} = \frac{\mathrm{F}}{\mathrm{A}} \times \frac{\ell}{\Delta\ell} \Rightarrow \mathrm{F} = \mathrm{YA} \cdot \frac{\Delta\ell}{\ell}$
On putting the values from question, $\mathrm{F} = \frac{5\times10^8\times25\times10^{-6}\times5\times10^{-2}}{10\times10^{-2}} = 6250$ N

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kinetic energy = potential energy of rubber $\frac{1}{2}\mathrm{mv}^2 = \frac{1}{2}\,\mathrm{F}\Delta\ell$

$\mathrm{v} = \sqrt{\frac{\mathrm{F}\Delta\ell}{\mathrm{m}}} = \sqrt{\frac{6250\times5\times10^{-2}}{5\times10^{-3}}} = \sqrt{62500}$

$= 250\ \mathrm{m/s}$

Q.13 Answer:

$1.77\times10^{-3}\ \mathrm{m}$

Solution:

Forces at the lowest point are gravitational force and centrifugal force

$\mathrm{F} = \mathrm{mg} + \mathrm{m}\omega^2\mathrm{r} = \mathrm{mg} + \mathrm{m}\omega^2\,\mathrm{L}$

$\Delta 1 = \frac{\mathrm{F}L}{\mathrm{YA}} = \frac{\left(\mathrm{mg}+\mathrm{m}\omega^2\,\mathrm{L}\right)\mathrm{L}}{\mathrm{Y}\times\pi\mathrm{r}^2}$

$\omega = 2\pi\mathrm{f} = 2\pi\times2 = 4\pi$

$\Delta 1 = \frac{\left(30+3\times16\pi^2\times1.5\right)\times1.5}{2\times10^{11}\times\pi\times(10^{-3})^2} = 1.77\times10^{-3}\ \mathrm{m}$

Q.14 Answer:

$1:1$

Solution:

$\mathrm{Y} = \frac{\mathrm{F}\ell}{\mathrm{A}\Delta\ell};\Rightarrow\quad \mathrm{F} = \frac{\mathrm{YA}\Delta\ell}{\ell}$

$\frac{\mathrm{F_1}}{\mathrm{F_2}} = \frac{\mathrm{A_1}}{\mathrm{A_2}}\times\frac{\mathrm{l_2}}{\mathrm{l_1}} = \frac{\mathrm{r_1^2}}{\mathrm{r_2^2}}\times\frac{\mathrm{l_2}}{\mathrm{l_1}} = \frac{4}{1}\times\frac{1}{4} = 1:1.$

Q.15 Answer:

$1.25\ \mathrm{mm}$

Solution:

$\mathrm{Y_c}\times(\Delta\mathrm{L_c}/\mathrm{L_c}) = \mathrm{Y_s}\times(\Delta\mathrm{L_s}/\mathrm{L_s})$

$\Rightarrow 1\times10^{11}\times\left(\frac{1\times10^{-3}}{1}\right) = 2\times10^{11}\times\left(\frac{\Delta\mathrm{L_s}}{0.5}\right)$

$\Delta\mathrm{L_s} = \frac{0.5\times10^{-3}}{2} = 0.25\ \mathrm{mm}$

Therefore, total extension of the composite wire

$= \Delta\mathrm{L_c} + \Delta\mathrm{L_s} = 1\ \mathrm{mm} + 0.25\ \mathrm{mm} = 1.25\ \mathrm{mm}$

Q.16 Answer:

$\frac{MgL}{2AY}$

Solution:

Let M be the mass of the rubber cord, L its length, A its cross-sectional area (assumed constant). Let us first find the elongation dl of an element AB of length dy at a distance y from the fixed end (Fig. 7.37). The force due to the weight of the cord is maximum at $P(y=0)$ and zero at $Q(y=L)$. Therefore, the force acting at the element is

$F = \frac{Mg}{L}(L-y)$ **(1)**

This force is responsible for the elongation of element AB. Now

or $Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{dl/dy} = \frac{Fdy}{Adl}$

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$dl = \frac{Fdy}{AY}$ **(2)**

Using Eq. (1) in Eq. (2), we get

$dl = \frac{Mg}{LAY}(L - y)dy$

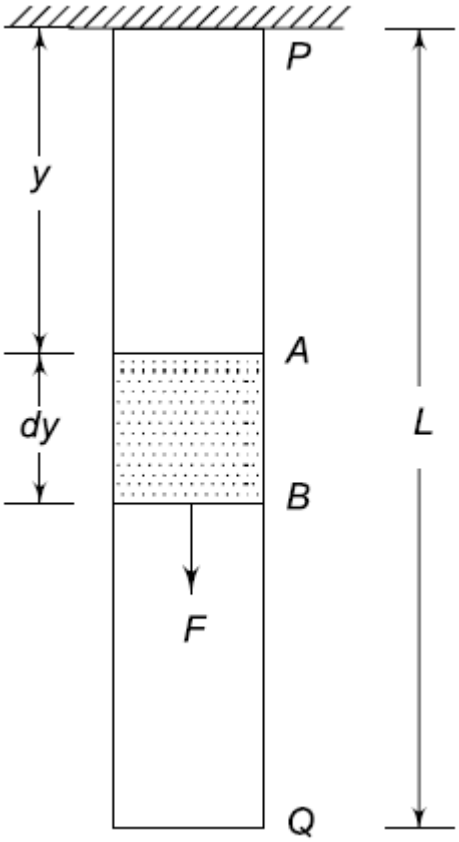
To obtain the total elongation l of the cord, we integrate from $y = 0$ to $y = L$. Thus

$$l = \int dl = \frac{Mg}{LAY}\int_0^L (L - y)dy$$

$$= \frac{Mg}{LAY}\left|Ly - \frac{y^2}{2}\right|_0^L$$

$$= \frac{Mg}{LAY}\left(L^2 - \frac{L^2}{2}\right) = \frac{MgL}{2AY}$$

The correct choice is (b).



Q.17 Answer:

24.00

Solution:

Slope of graph $= \frac{l}{F} = \frac{1.2 \times 10^{-1}}{x}$

$Y = \frac{FL}{Al} \Rightarrow \frac{l}{F} = \frac{L}{YA}$

$\therefore \frac{1.2 \times 10^{-4}}{x} = \frac{1}{2 \times 10^{11} \times 10^{-4}}$

$\therefore \quad x = 2.4 \times 10 = 24\text{N}$

Q.18 Answer:

Half of stress $\times$ strain

Solution:

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$U =$ energy per unit volume $= \frac{1}{2} \times$ stress $\times$ strain

$= \frac{1}{2} \frac{F}{A} \times \frac{\Delta l}{l} = \frac{1}{2} \times Y \times \frac{\Delta l^2}{l}$

Q.19 Answer:

0.02

Solution:

Work done in stretching the wire,

$W = \frac{1}{2} \times \quad load \quad \times extension = \frac{1}{2} F \times l$

Substituting for $l = \frac{FL}{\pi r^2 Y}$

$\therefore \quad W = \frac{1}{2} F \times \frac{FL}{\pi r^2 Y} = \frac{F^2 L}{2\pi r^2 Y}$

75% of work done in stretching the wire is converted into heat energy.

∴If $\Delta\theta$ **is rise in temperature, then**

$0.75W = mc\Delta\theta = \pi r^2 L \rho c \Delta\theta$

$\Rightarrow \pi r^2 L \rho c \Delta\theta = \frac{0.75 F^2 L}{2\pi r^2 Y}$

$\therefore \quad \Delta\theta = \frac{0.75 F^2}{2\pi^2 r^4 Y \rho c}$

$= \frac{0.75 \times (40 \times 9.8)^2}{2\pi^2 \times 10^{-3^4} \times 7.5 \times 10^{10} \times 8 \times 10^3 \times 490}$

$= \frac{0.75 \times 40 \times 40 \times 9.8^2}{2 \times (9.8) \times 10^{-12} \times 7.5 \times 10^{10} \times 8 \times 10^3 \times 490}$

$\therefore \quad \Delta\theta = \frac{100 \times 9.8}{49 \times 10^3} = \frac{1}{50} = 0.02K$

Q.20 Answer:

8

Solution:

Let L be original length of string and k be force constant of string. New final length = initial length + elongation

$L' = L + \frac{F}{K}$

When tension of $3 \quad N$ **is applied,**

$x = L + \frac{3}{k}$ **...(i)**

When tension of $5N$ **is applied,**

$y = L + \frac{5}{k}$ **....(ii)**

Subtracting equation (ii) from (i),

$x - y = \frac{3}{k} - \frac{5}{k}$

$\therefore \quad k = \frac{-2}{x - y} = \frac{2}{y - x}$

Substituting value of k **in equation (i),**

$x = L + \frac{3}{2/y - x} = L + \frac{3(y - x)}{2}$

$\therefore \quad L = x - \frac{3(y - x)}{2} = \frac{2x - 3y + 3x}{2} = \frac{5x - 3y}{2}$

Let T be tension when length is $2.5 \quad y - 1.5 \quad x$

$\therefore \quad 2.5y - 1.5x = L + \frac{T}{k}$

$2.5y - 1.5x = \frac{5x - 3y}{2} + \frac{T}{(2/y - x)}$

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$2.5y - 1.5x = \frac{5x - 3y}{2} + \frac{(y - x)T}{2}$

$\therefore \quad 5y - 3x = 5x - 3y + (y - x)T$

$\therefore \quad 8y - 8x = (y - x)T$

$\therefore \quad 8(y - x) = (y - x)T \Rightarrow T = 8N$

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